Abhishek Singh

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Education

Central University of Chhattisgarh, Bilaspur Dec 2022 – Apr 2026
Bachelor of Technology (Major in Chemical, Minor in AI) **CGPA: 8.1**

Publications

- **Diagnosing and Mitigating the Intonation Bottleneck in Semantic Audio Tokens.** *Submitted to ACL ARR 2026.* Proposed **Hybrid MCST**, a contrastive adapter raising intonation accuracy from 37.2% to **66.9%**.
- **Towards Optimal Indic Code-Switching SLAM-ASR: An Empirical Search for the Best Encoder-Decoder Synergy.** *Submitted to ACL ARR 2026.* Compared **6 speech encoders and 3 LLMs** in SLAM-LLM across Hindi, Tamil, Kannada, and Marathi.

Experience

Speech Lab, Indian Institute of Technology (IIT) Madras
May 2026 – Present
Research Staff *Onsite*

- Worked on ASR research across **15 Indian languages**; trained an Indic-English code-switch model (SLAM-LLM: data2vec-AQC + Gemma-3-4B-IT with LoRA) on a ∼5000hr corpus using DeepSpeed ZeRO-2 on 4 GPUs.
- Fine-tuned **Sarvam, Qwen, Gemma, and Airavata** models on Indian-language and code-switch data.
- Built a VAD-based longform ASR pipeline with rolling-context conditioning; fixed five hallucination bugs (repeat-loops, timestamp leakage, script substitution, etc.).
- Outperformed Meta's Omnilingual ASR-7B on 8/9 Hindi/code-switch benchmarks with a ∼30M-parameter fine-tune.
- Open-sourced the model and training corpus on Hugging Face (Apache 2.0) with documentation and benchmarks.

GlofiHub
Jun 2026 – Aug 2026
Founding Engineer *Remote*

- Built **AI-driven education products** for GlofiHub, an AI-integrated group offering overseas/MBBS counseling and digital services **globally**.
- Developed the **mobile app, web platform, backend, and database** end to end, from design to deployment.
- Led **AI/ML features** and **platform architecture** for student guidance and counseling workflows.

Indian Institute of Science (IISc), Bangalore (TANUH.ai)
Nov 2025 – May 2026
Quantum Machine Learning Research Intern

- Designed a **QPNN model** for time-series forecasting on non-stationary cloud workloads, outperforming GRU baselines.
- Built a memory-efficient streaming training pipeline (**under 2 GB RAM**) for large-scale Alibaba cluster datasets.
- Achieved a **68–73% reduction in MSE** across 1–20 min forecasting horizons through model optimization and tuning.
- Developed and evaluated an end-to-end ML pipeline for preprocessing, training, and real-world deployment.

Indian Institute of Technology (IIT) Hyderabad
May 2025 – Jun 2025
AI/ML Research Intern

- Developed object detection algorithms for **self-driving cars** using the **YOLO family** of models with PyTorch and OpenCV, achieving **84% accuracy** under real-world conditions.
- Built an end-to-end perception pipeline integrating object and lane detection with multi-sensor fusion (LiDAR, GPS) for real-time driving decisions.
- Improved detection performance by **15%** through hyperparameter tuning, optimization, and systematic evaluation.
- Validated model robustness across diverse lighting and environmental conditions for reliable deployment.

Projects

Sutra-1.3B — 1.32B-Parameter Mixture-of-Experts (MoE) LLM Live Demo | Hugging Face

- Pretrained a **1.32B-parameter MoE LLM** (0.28B active/token) from scratch on **18B tokens** across 4 GPUs using pure PyTorch, with a custom 48K BPE tokenizer and data pipeline.
- Implemented **Multi-head Latent Attention (MLA)** and auxiliary-loss-free expert routing inspired by DeepSeek, reducing KV cache by **7.1**× (64 KB → 9 KB per token).
- Aligned the model using **SFT and DPO** on 300K samples, achieving **65.6% PIQA** and **45.0% ARC-Easy**.
- Deployed a **24/7 FastAPI + Next.js** inference system with Wikipedia retrieval, web search, and INT8 quantization, reducing model size by **1.86**×.

VidyaAI — AI-Powered Multilingual Animated Lesson Platform Source Code | Deployed Link

- Built a full-stack **Next.js 14 + TypeScript** platform that turns any topic into a **4–5 min animated lesson** with synchronized voice narration in **22 Indian languages**.
- Developed an **LLM + RAG-driven pipeline** (script, scenes, Manim export) with a live **SVG scene renderer**, and a library of **40+ lessons** across Math, Science, and CS.
- Implemented **NextAuth (Google + guest)**, Prisma + Postgres, streaks, and watch history on **Vercel + Render**.

ShrutamMT-v1 (Multilingual Machine Translation) Hugging Face

- Trained a **148.5M-parameter Transformer from scratch** for 11 Indic languages → English using **49.2M+ Samanantar sentence pairs**.
- Implemented the complete **Encoder-Decoder Transformer in PyTorch**, including Multi-Head Attention, positional encoding, residual connections, label smoothing, and beam search.
- Achieved **29.76 BLEU / 52.61 chrF** after **400K training steps**, with 40.91 BLEU on Bengali.
- Used **64K joint BPE, tied embeddings, language tags, Noam LR scheduling**, and checkpoint averaging.

Technical Skills

- **Programming Languages:** Python, SQL, Data Structures & Algorithms, Object-Oriented Programming
- **AI / ML:** Machine Learning, Deep Learning, Computer Vision, NLP, Time-Series Forecasting,
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-Learn, OpenCV, JAx, Selenium
- **Web & Backend Development:** React.js, React Native, Electron.js, Node.js, FastAPI, Flask, REST APIs, flutter, Kotlin, Shorebird.
- **System Design:** HLD & LLD (Basics)
- **Tools:** Git, GitHub, Linux, Jupyter, Google Colab
- **AI Automation:** Langchain, Langraph, LangSmith, Langflow, Crewai, Agno
- **Cloud & DevOps:** Microsoft Azure (Basics), Docker, CI/CD (Basics)
- **Databases:** MongoDB, MySQL, PostgreSQL, Supabase, ChromaDB

Hackathon & Achievements

- **Smart India Hackathon (SIH) 2025 – Winner, Rank 1 among 600K+ participants:** built **Prashikshak**, a disaster-management training platform for NDMA/SDMAs **(Ministry of Home Affairs)**. [Deployed Link]
- **Hacktivate 2025 & National Innovation Award (NIA) 2024 – Winner (both):** built **AgroSmart**, an **IoT-based smart farming system** that reads field parameters (temperature, soil, etc.) to recommend **fertilizer and crops**. [Deployed Link]
- **GGV Ideathon 2024 – Winner, Rank 1:** proposed a **rental rooms platform for students** to find and book affordable rooms.
- **BNI Shark Tank 2025 – 2nd Winner:** built **Labour-Connect**, a service marketplace to find and book **verified local workers**, using **Flutter, FastAPI, and a React admin panel**.
- **Grant Recipient – CBDE:** received a grant for the **AgroSmart** IoT-based smart farming project.
- **Hack For Impact (IIT Delhi) – Top 9:** presented **AgroSmart**, the IoT-based smart farming project.